TASK SPECIFICATION: P3-05 PHASE P3 — FRONTEND ARCHITECTURE (REACT 19 + TYPESCRIPT)

Shared UI Component Kit (Modal, Toast, Badge, Button, SaudiPlate, QR)

Git Branch: `feature/P3-05-ui-kit` · Target Pull Request: feature/P3-05-ui-kit

Core Objective & Technical Scope:
Reusable atomic component library: Modal with focus-trap, Toast notification manager, Badge, Button, FormField, SaudiPlate renderer, and QR code generator.

Priority Level	Medium	Complexity Rating	`M`
Execution Status	Backlog 📋	Target PR / Commit	`feature/P3-05-ui-kit`
Upstream Dependencies	P3-01		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Implement atomic components matching design system token definitions.
2. Implement accessible Modal component with focus trap and escape key listeners.
3. Implement SaudiPlate component rendering official Saudi plate layout with Arabic lettering.
4. Perform automated accessibility audit using axe-core.

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `frontend/src/shared/ui/*`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- All interactive components fully operable via keyboard.
- Zero accessibility violations reported by axe-core.

VERIFICATION & TEST COVERAGE STRATEGY

- Component unit tests
- Axe accessibility compliance tests